

overview

the module, the project, and where everything belongs · prof. dr. nicolas meseth

- 1 the goal
- 2 one question, four questions
- 3 five challenges
- 4 how we work

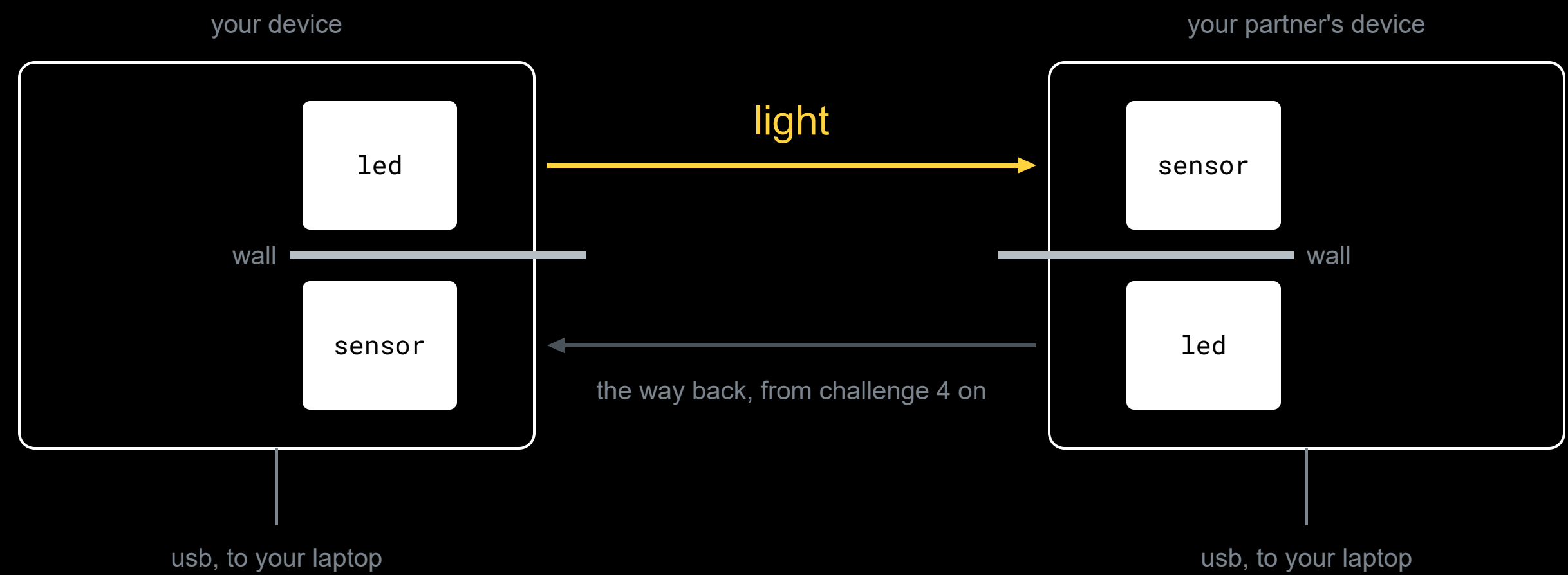
part 1

the goal

there is a photo
on this laptop.
how does it get
to the other one?



the device



a lamp, an eye, and a wall between them. two of them face each other, so each can send and receive.

why a lamp

everything interesting in a computer is invisible.

it happens inside chips, in millionths of a second, behind software somebody else wrote.

this device does the same thing a few times a second, on your desk.

you can put your hand in the way and watch what goes wrong.

everything this module wants to teach you can be seen on that table.

what this module is about

how can we solve complex problems with computers?

one question. every week is a piece of the answer.

part 2

one question, four questions

the map

how can we solve complex problems with computers?

the map

how can we solve complex problems with computers?

how do computers
represent information?

how do computers
store information?

how do computers
transfer information?

how do computers
process information?

how can we solve complex problems with computers?

how do computers
represent information?

making bits

how do computers
store information?

storing bits

how do computers
transfer information?

sending bits

how do computers
process information?

processing bits

how can we solve complex problems with computers?

how do computers
represent information?

making bits

analog and digital

symbols and information

number systems

code systems

how do computers
store information?

storing bits

memory

files

how do computers
transfer information?

sending bits

signal and noise

sampling and sync

protocols

errors and redundancy

throughput and limits

how do computers
process information?

processing bits

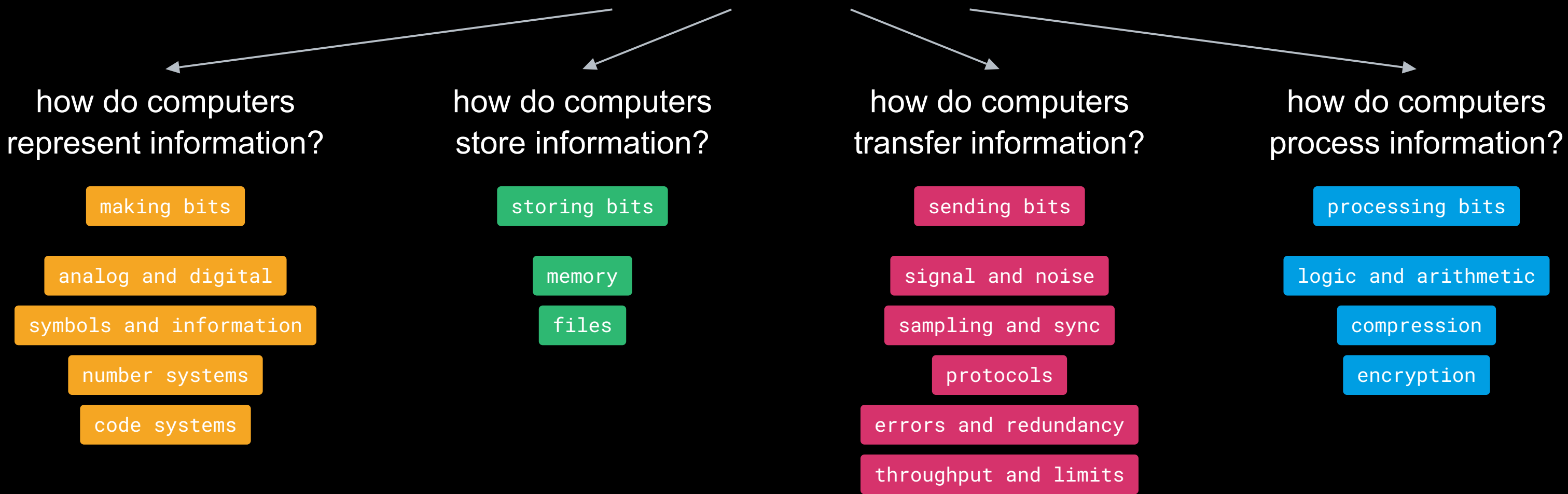
logic and arithmetic

compression

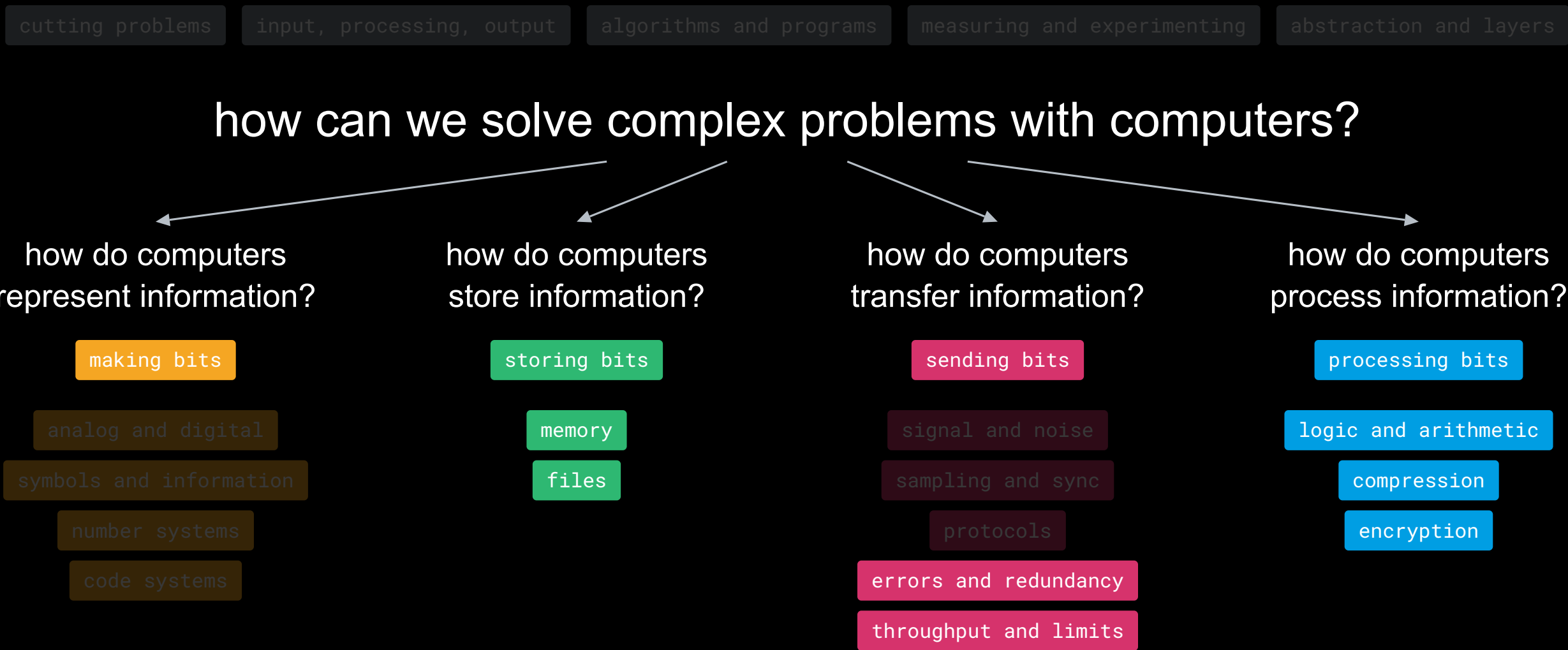
encryption

- cutting problems
- input, processing, output
- algorithms and programs
- measuring and experimenting
- abstraction and layers

how can we solve complex problems with computers?



the project on the map



challenge 4: the packet

your device touches all four questions. that is why it is the project.

the four questions, in one line each

represent to a computer, your photo is a list of numbers, and the numbers are states of something physical

store the photo has to sit somewhere before it is sent, and somewhere after it arrives.

transfer the receiver does not know when a letter begins. the light it sees is never the light that was sent.

process somebody has to turn the numbers back into a picture, and check whether they arrived correctly.

what the exam is about

the concepts.

the project is the material on which you explain them.

whoever passes every challenge and cannot say why digital transmission tolerates noise does not pass the exam.

the map lives on the website

LiFi Project

Concepts > The Map

Concepts

The device you build is not the goal. It is the object on which a set of concepts becomes visible that would otherwise stay invisible. These pages collect them.

Every concept meets you at a particular point in the project, usually as a problem before it appears as a term. That is deliberate: you are meant to notice first that something does not work, and to learn afterwards what it is called and why it behaves that way.

Every page starts with a summary and the questions it is about. Then the concept is explained in full, with its diagrams. For looking things up again, the slides from the session follow, and later some further reading.

Take this part seriously: **the concepts are what the exam is about**. Your project is the material on which you explain them, but it does not replace understanding them. Whoever has passed every challenge and still cannot say why digital transmission tolerates noise does not pass the exam.

One question, four questions

The module has one big question: how can we solve complex problems with computers? Four questions about information sit underneath it, and every concept on these pages belongs to one of them. Above all four lie the ways of working that you need everywhere: cutting problems, the input-processing-output model, measuring, and thinking in layers. The colours on this map are the same on the slides: whenever a session works on one of the four questions, its colour tells you which.

cutting problems

input, processing, output

algorithms and programs

measuring and experimenting

abstraction and layers

how can we solve complex problems with computers?

part 3

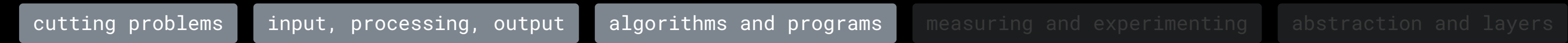
five challenges

one new difficulty at a time

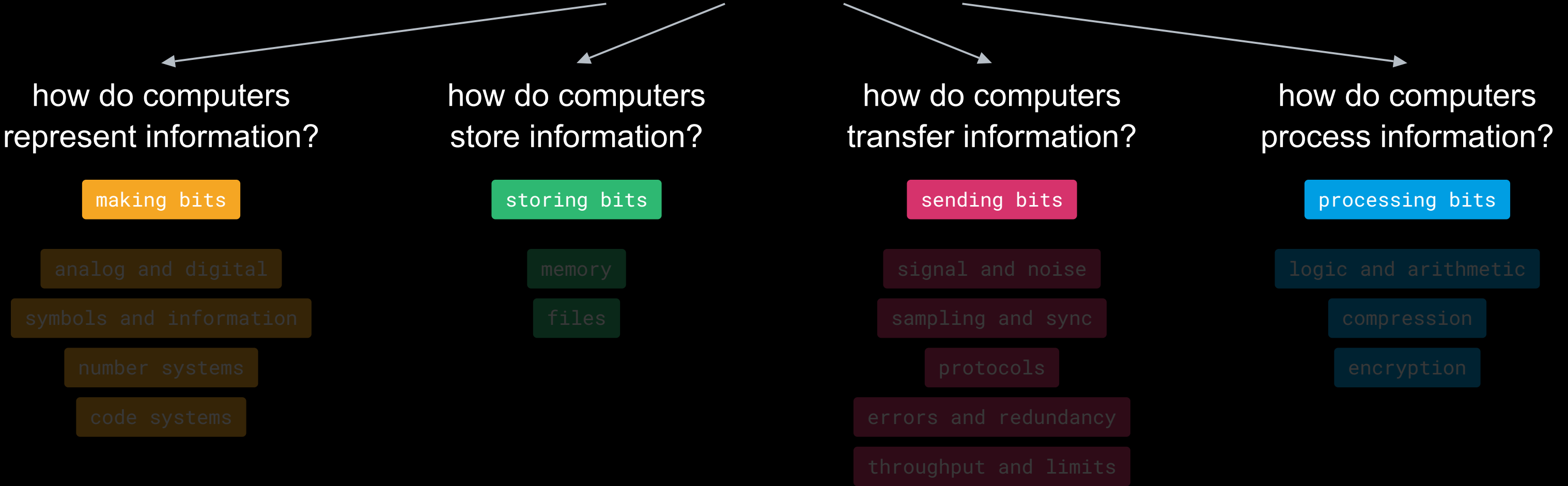
- | | | |
|---|--------------|--|
| 0 | the spark | can you bring your device to life? |
| 1 | the alphabet | how many signals can your receiver tell apart? |
| 2 | the word | can you send a whole word, and how fast? |
| 3 | the listener | can the receiver find the start on its own? |
| 4 | the packet | can you send a real file, correct and fast? |

each challenge adds exactly one new difficulty. everything else stays as you know it.

where each challenge sits on the map

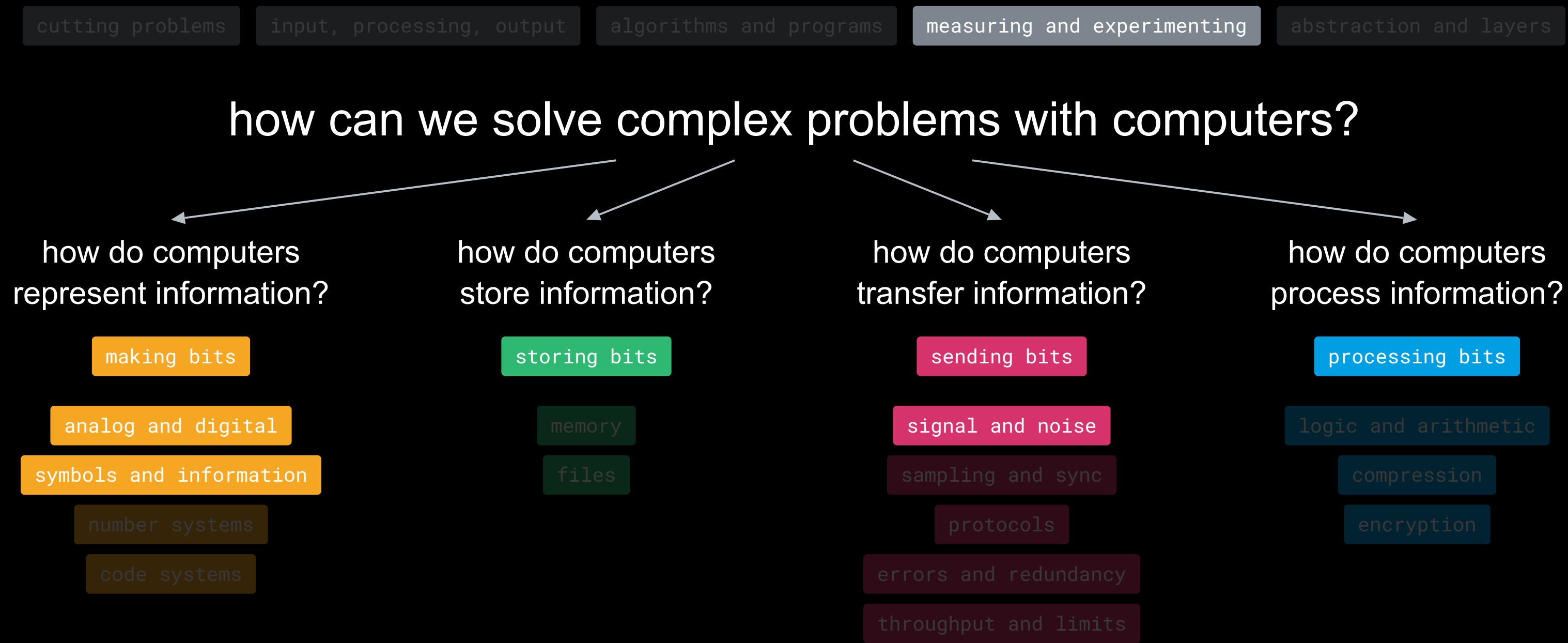


how can we solve complex problems with computers?



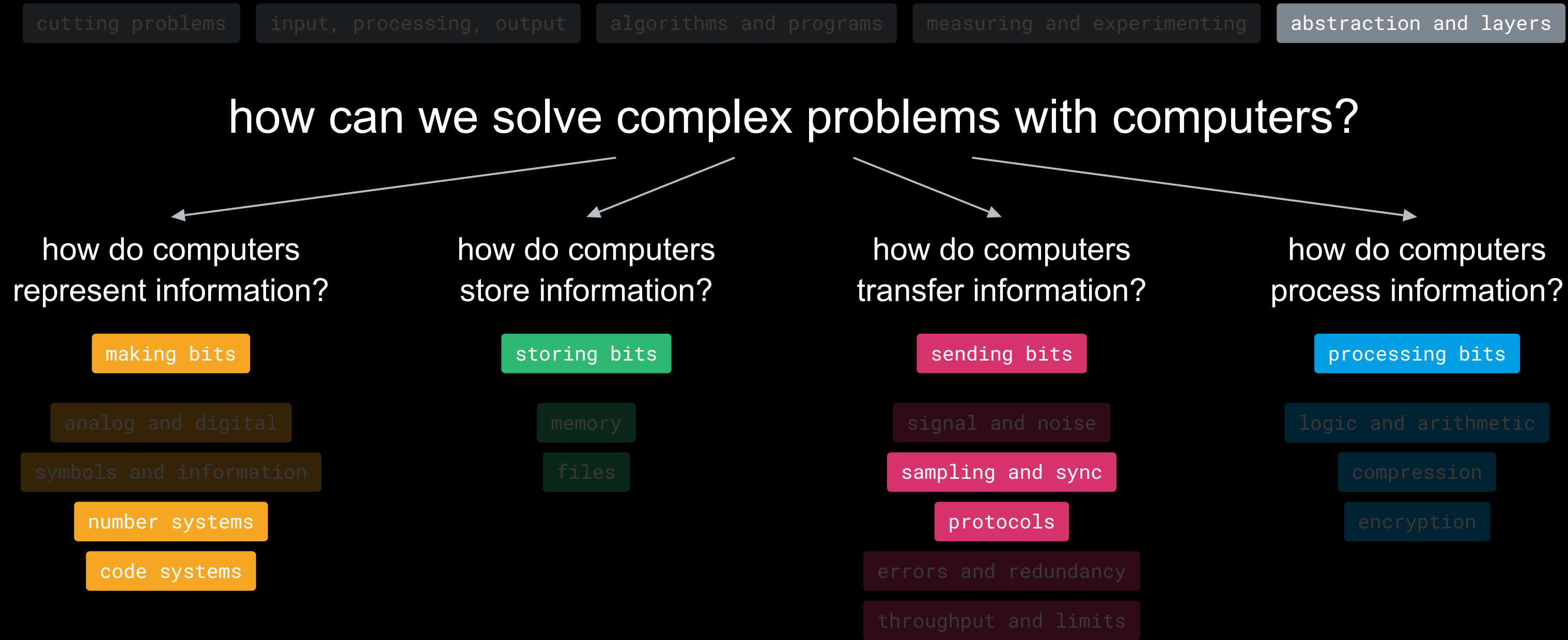
challenge 0: the spark

where each challenge sits on the map



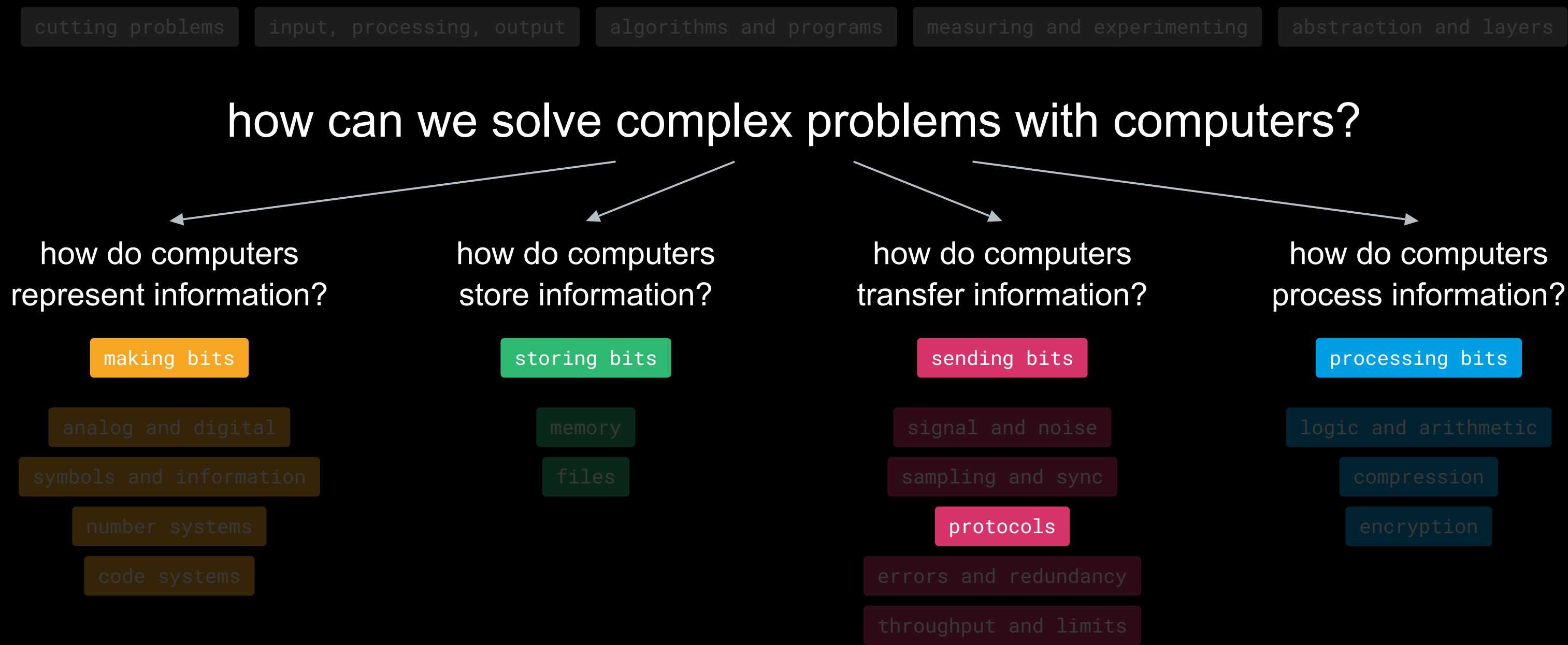
challenge 1: the alphabet

where each challenge sits on the map



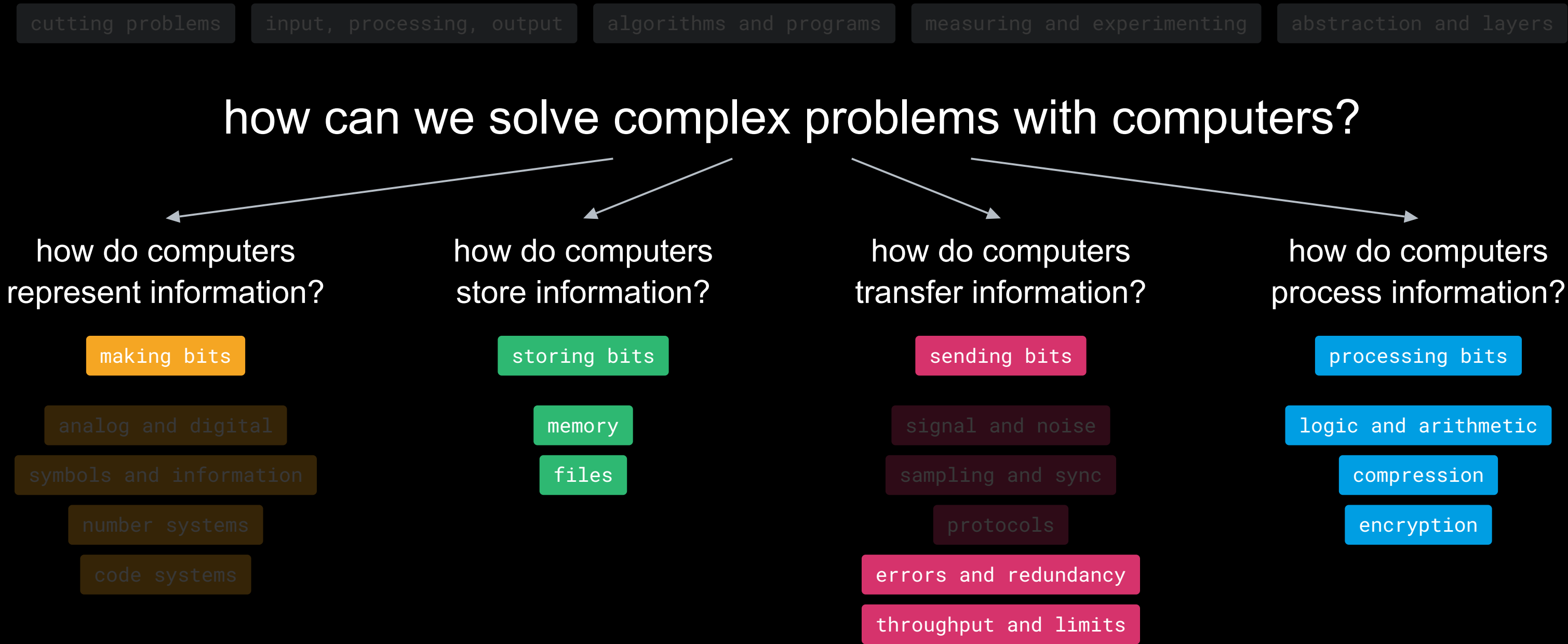
challenge 2: the word

where each challenge sits on the map



challenge 3: the listener

where each challenge sits on the map



challenge 4: the packet

what every challenge produces

measurement logs

what did you change,
what did you keep the same,
and what came out?

a description of your solution

enough for another team
to rebuild it. from challenge 3 on,
a full protocol specification.

a log of mistakes

where the assistant suggested
something, and a measurement
proved it wrong.

what counts

the ranking in the competition does not count towards your grade.
passing every challenge is what admits you to the exam.
the exam is about the concepts, one person at a time.

part 4

how we work

two people, two devices, one link

one device each, so between the two of you, you own both ends.
whoever builds only the sender has to guess what the receiver sees. you can look.
python, vs code, and an ai assistant at your side from day one.

the lamp has the last word

your assistant knows nothing
about your device.

it will tell you that eight colours are a good alphabet. is that right?
there is one honest way to find out.

measure before, measure after, and write down what the assistant got wrong.

concept demos

small interactive apps, one idea each. play with them; the concepts are from the lifi project.

search demos... (press / to focus)

all

input, processing & output

measurement & experiments

symbols & information

analog & digital

signal & noise

sampling & synchronization

protocols

throughput & limits

number systems

code systems

files

logic & arithmetic

compression

encryption

18 demos



input, processing & output

the ipo test lab

change sensor values and the classification rule, compare actual with expected output, and watch the test pass or fail.



measurement & experiments

the noisy sensor

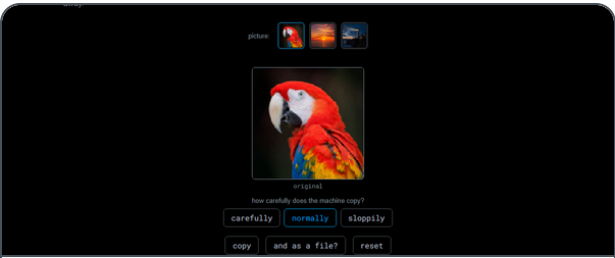
send a symbol over a noisy link and let the same readings decide it three times: alone, averaged over five, averaged over ten. fewer wrong decisions, less speed.



symbols & information

question game

find the secret face among sixteen: every answer shrinks the crowd, the shrinking is measured in bits, and the halving strategy shows why four questions are enough.



analog & digital



analog & digital



analog & digital

how we work

today

- 1 one question, four questions: represent, store, transfer, process.
- 2 nineteen concepts, five colours, one map. you will see it again.
- 3 five challenges, one new difficulty each.
- 4 the exam is about the concepts. the project is where you meet them.
- 5 the lamp has the last word.

next: install the software, read the hardware page, start challenge 0.

by the end of term,
a picture crosses
that gap.

